
SSM Optimization

Weekly Meeting

27/02/26

Miscellaneous

- I have done some debugging to check the Falcon Mamba's architecture step by step.
- I did run experiments with Mamba-2.8B, Qwen3-4B, Qwen3-0.6B, Qwen3-8B
- Debugging Falcon-H1-1.5B-Base, as it gives OoM.

Evaluation on Wikitext

Rescaling by 2048 to be applied: instead of average KL per token, I aggregated KL of the whole sequence

Model	Quantized	Wikitext2				Memory (GB) [SINQ]
		SINQ		RTN		
		Perplexity	KL divergence	Perplexity	KL divergence	
Falcon Mamba 7B		5.203	-	-	-	14.56
Falcon Mamba 7B	 (8 bits)	5.159	18.946	5.158	18.965	8.28
Falcon Mamba 7B	 (4 bits)	5.378	99.482	5.397	103.422	4.95
Falcon Mamba 7B	 (3 bits)	5.987	331.053	6.035	349.105	4.22
Falcon Mamba 7B	 (2 bits)	18.514	2633.312	36.174	3995.665	3.24
Qwen3-8B		9.727	-	-	-	16.38
Qwen3-8B	 (8 bits)	9.716	1.778	9.719	2.056	9.90
Qwen3-8B	 (4 bits)	9.838	127.373	9.974	160.756	6.48
Qwen3-8B	 (3 bits)	11.749	579.109	12.391	712.955	5.73
Qwen3-8B	 (2 bits)	8573.92	14971.098	14965.382	16119.065	4.71

Evaluation on Wikitext

Rescaling by 2048 to be applied: instead of average KL per token, I aggregated KL of the whole sequence

Wikitext2						
Model	Quantized	SINQ		RTN		Memory (GB) [SINQ]
		Perplexity	KL divergence	Perplexity	KL divergence	
Mamba-1.4B		10.761	-	-	-	2.75
Mamba-1.4B	 (8 bits)	10.753	1.331	10.753	1.266	1.56
Mamba-1.4B	 (4 bits)	11.094	64.104	11.225	79.748	0.93
Mamba-1.4B	 (3 bits)	12.627	326.925	12.870	375.701	0.81
Mamba-1.4B	 (2 bits)	64.315	3626.683	88.682	4263.727	0.613
Qwen3-1.7B		16.714	-	-	-	3.44
Qwen3-1.7B	 (8 bits)	16.645	2.704	16.66	4.22	2.12
Qwen3-1.7B	 (4 bits)	17.189	229.498	18.75	375.382	1.42
Qwen3-1.7B	 (3 bits)	23.268	1107.034	32.498	1814.610	1.29
Qwen3-1.7B	 (2 bits)	25379.449	16381.184	202540.781	21139.056	1.07

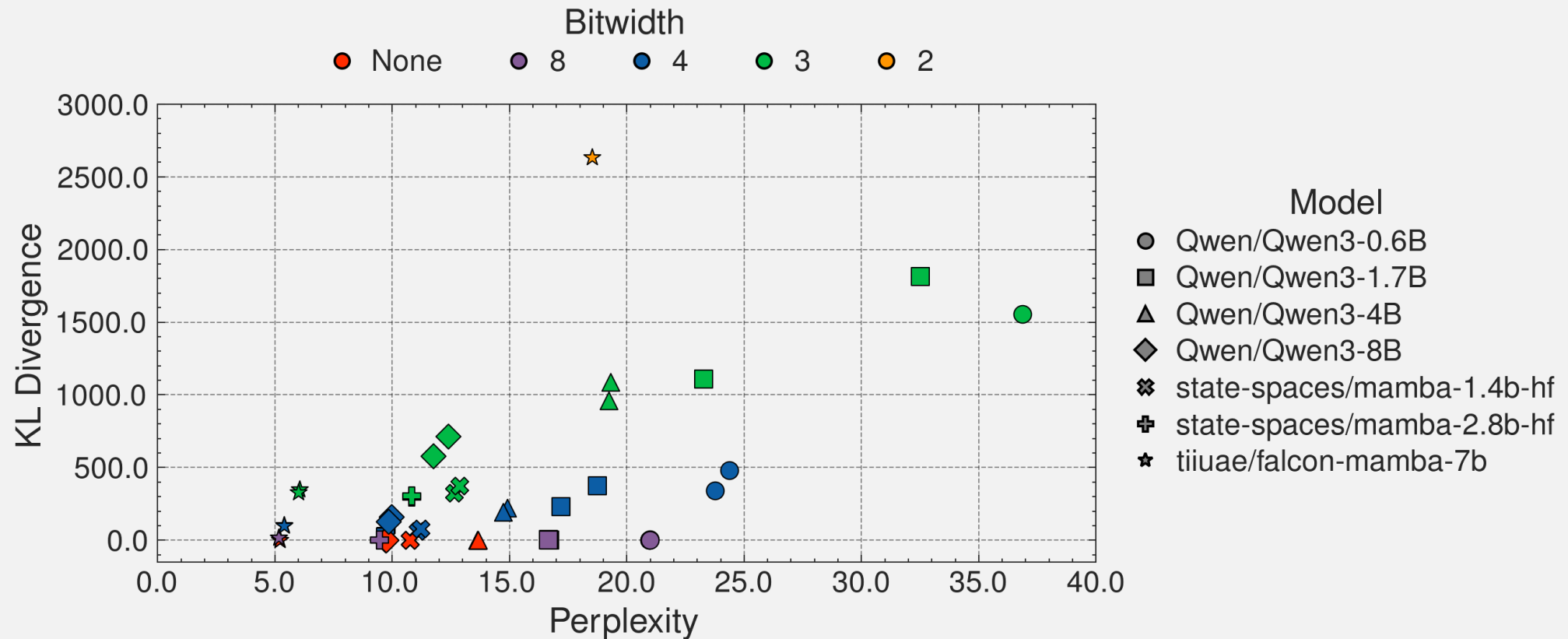
Evaluation on Wikitext

Rescaling by 2048 to be applied: instead of average KL per token, I aggregated KL of the whole sequence

Model	Quantized	Wikitext2				Memory (GB) [SINQ]
		SINQ		RTN		
		Perplexity	KL divergence	Perplexity	KL divergence	
Mamba-2.8B		9.462	-	-	-	5.621
Mamba-2.8B	 (8 bits)	9.450	3.230	9.450	3.228	3.07
Mamba-2.8B	 (4 bits)	9.713	61.813	9.724	65.213	1.76
Mamba-2.8B	 (3 bits)	10.818	302.866	10.822	295.758	1.50
Mamba-2.8B	 (2 bits)	72.718	4181.002	65.736	3985.892	1.11
Qwen3-4B		13.666	-	-	-	8.10
Qwen3-4B	 (8 bits)	13.628	2.109	13.623	2.301	4.67
Qwen3-4B	 (4 bits)	14.721	195.475	14.908	224.608	2.84
Qwen3-4B	 (3 bits)	19.220	959.884	19.311	1088.881	2.47
Qwen3-4B	 (2 bits)	72940.484	18836.552	82132.156	19461.636	1.93

Evaluation on Wikitext

Spearman correlation: 0.658



Next steps

- Get the related techniques' code working (Quamba, Quamba2)
- Run experiments also on other datasets (C4 + Im-eval)
- Sensitivity analysis within each Mamba block?